and microsite, both of which contain GHG climate-related information." This company estimates the cost of producing voluntary climate-related disclosures to be less than 5 percent of its total SEC compliance-related costs. The remaining company, a large-cap industrial manufacturing company, reported that the combined cost of producing its first TCFD, SASB, and GRI disclosures amounted to between $250,000 and $350,000 (without providing a breakdown of component costs), while the cost of responding to its first CDP questionnaire was less than $50,000. This company reported that it also spent $400,000 annually for third-party auditors and consultants that provide support in the company's climate disclosure efforts.

Another commenter provided the results of a survey, conducted from February to March 2022, of corporate issuers and institutional investors ("ERM survey").[2988] The results reflect the responses of 39 issuers, of which 29 were LAFs.[2989] The ERM survey presents issuers' average annual costs in seven categories: GHG analysis and/or disclosures ($237,000);[2990] climate scenario analysis and/or disclosures ($154,000);[2991] additional climate-related analysis and/or

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2988 *See* ERM survey, *supra* note 2670. The 39 issuers included the following industries: healthcare and pharmaceuticals; financials, insurance, and professional services; consumer discretionary products; communication services; transportation, construction, and industrials; consumer staples; oil, gas, and energy; utilities; real estate; metals, plastics, and other raw material; and information technology.

2989 *See id.* Respondent market capitalizations ranged from less than $300 million to more than $200 billion, with the highest proportion of respondents (34%) having a market capitalization between $10 billion and $50 billion.

2990 This survey category includes all costs relating to the development of GHG inventories with analysis and disclosure of Scope 1, Scope 2, and/or Scope 3 emissions.

2991 This survey category includes all costs to issuers related to conducting assessments of the impact of climate risks in the short-, medium-, or long-term using scenario analysis as well as TCFD/CDP disclosure of risks and opportunities. Respondents were asked to exclude from this category any costs that they included in their costs of GHG emissions analysis and disclosures.

disclosures ($130,000);[2992] internal climate risk management controls ($148,000);[2993] proxy responses to climate related proposals ($80,000); assurance/audits related to climate ($82,000); and other climate-related disclosure costs not covered by the previous six categories ($76,000).

We also reviewed annual cost estimates associated with existing climate-related disclosure policies in the U.K. In 2021, the U.K. Financial Conduct Authority ("FCA") adopted a comply-or-explain disclosure rule ("FCA rule"), which originally applied only to commercial companies with a U.K. premium listing[2994] but, effective 2022, was subsequently expanded to include issuers of standard listed shares.[2995] The U.K. Department for Business, Energy, and Industrial Strategy ("BEIS") adopted a similar—albeit mandatory—disclosure rule ("BEIS

2992 This survey category includes additional voluntary climate-related analyses and disclosures for processes largely disconnected from current and proposed climate-related disclosures such as outreach, engagement, and management.

2993 This survey category includes costs for internal climate risk management controls, namely the costs related to integrating climate risk into enterprise risk management, oversight at the board level, strategic planning, internal audit, and other fundamental business processes. In addition, this category includes issuer costs related to climate-related data collection and aggregation, including IT costs and staff time; internal review of climate-related data collection by management, board committees, and the board; in-house counsel drafting; and review by outside counsel.

2994 *See* FCA, Policy Statement PS20/17, *Proposals to Enhance Climate-related Disclosures by Listed Issuers and Clarification of Existing Disclosure Obligations* (Dec. 2020), available at https://www.fca.org.uk/publication/policy/ps20-17.pdf. This document states that the rule would apply to 480 companies that have a premium listing. A premium listed company is a company listed on the London Stock Exchange that is subject to more stringent compliance and disclosure requirements in addition to the minimum standards outlined in the UK provisions that implemented the EU Consolidated Admissions and Reporting Directive (CARD) and the EU Transparency Directive.

2995 *See* FCA, Consultation Paper CP21/18, *Enhancing Climate-related Disclosures by Standard Listed Companies and Seeking Views on ESG Topics in Capital Markets* (June 2021), available at https://www.fca.org.uk/publication/consultation/cp21-18.pdf. Cost estimates of the FCA rule are sourced from this document. *See also* FCA, Policy Statement PS21/23, *Enhancing Climate-related Disclosures by Standard Listed Companies* (Dec. 2021). This rule applies to 244 issuers: 148 issuers of standard listed equity shares as well as 96 additional issuers (i.e., standard listed issuers of Global Depository Receipts and standard listed issuers of shares other than equity shares, excluding standard listed investment entities and shell companies). A standard listed company is a company listed on the London Stock Exchange that is subject to the minimum standards outlined in the UK provisions that implemented the EU Consolidated Admissions and Reporting Directive (CARD) and the EU Transparency Directive.

rule"), also effective 2022,[2996] that was previously used to inform the Commission's cost estimates of the proposed rules.[2997] The BEIS rule generally applies to companies that have over 500 employees and/or a turnover of more than £500 million.[2998] Both rules exhibit significant overlap as they are both largely based on the TCFD framework's major components, including disclosure on governance, strategy, and risk management, all of which have similar counterparts in the final rules. Both UK rules also include scenario analysis and metrics and targets; however, because undertaking these activities is not required under the final rules (only their disclosure in specific circumstances), we focus on the cost estimates of the other components that are more relevant to the final rules.[2999]

One-time implementation costs—which consist of "familiarization costs" and "legal review"—are estimated to be $19,543[3000] by the BEIS and $15,147 by the FCA. The BEIS rule

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2996 U.K. Final Stage Impact Assessment, Mandating Climate-related Financial Disclosures By Publicly Quoted Companies, Large Private Companies and Limited Liability Partnerships (LLPs) (2021), available at https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/1055931/tcfd-final-stage-ia.pdf ("BEIS Final Stage Impact Assessment"). Cost estimates of the BEIS rule (with the exception of familiarization costs) are sourced from this document.

2997 *See* Proposing Release, section IV.C.2.a, which also reviews the BEIS rule (referred to as the "UK Impact Assessment" in the Proposing Release). The estimated costs of the BEIS rule, as outlined in the following paragraphs, are the same as those presented in the proposed rules, with the exception of applying an updated exchange rate to convert the costs from GBP to USD.

2998 Specifically, the BEIS rule applies to relevant Public Interest Entities ("PIEs"), including UK Premium and Standard listed companies with over 500 employees, UK registered companies with securities admitted to the Alternative Investment Market with more than 500 employees, limited liability partnerships with over 500 employees and a turnover of over £500 million, and UK registered companies that are not included in the categories above and that have over 500 employees and a turnover of over £500 million.

2999 The following cost estimates from the BEIS and FCA reflect internal labor costs with the assumption that affected entities have no pre-existing climate-related disclosure practices that fulfill the stated requirements. The costs are converted from GBP to USD using the 2022 average exchange rate of $1.2369 USD/GBP.

3000 The familiarization cost component is sourced from the BEIS Consultation Stage Impact Assessment (as opposed to Final Stage), which assumes that scenario analysis requirements are not part of the familiarization process, and thus may be a relatively better representation of the corresponding cost with respect to the final rules. The familiarization cost is estimated to be £12,600. *See* BEIS Final Stage Impact Assessment2996. The other initial cost related to legal review (£3,200), as outlined in the BEIS Final Stage Impact Assessment, is added to obtain £15,800 ($19,543). *See supra* note 2996.

presents first-year cost estimates of complying with climate-related disclosures associated with Governance ($11,256),[3001] Strategy ($16,080),[3002] and Risk Management ($13,359),[3003] for a combined total of $40,694, which is assumed to remain the same in subsequent years. This contrasts with the FCA's corresponding first-year costs of $183,028 and ongoing costs of $86,270 for larger issuers.[3004] The FCA rule also estimates costs for small and medium-sized issuers, with corresponding costs of $137,271 in the first year and $64,702 in subsequent years.

These estimates from the FCA and BEIS rules help to inform our assessment of the compliance costs of similar provisions of the final rules amending Regulation S-K, as our approach for these provisions is based, in part, on the TCFD recommendations. However, it is important to note that these estimates are intended to reflect compliance costs of the typical company within the designated sample of affected entities and are conditional upon several assumptions regarding the number of required staff, the rank or title of the staff, the required labor hours, and local wage data. Actual costs can vary significantly depending on company characteristics, such as company size, industry, business model, the complexity of the company's corporate structure, existing climate-related disclosure practices, and internal expertise, etc.

3001 Governance costs include the ongoing cost to those in scope to implement, document and disclose governance of their climate related risks and opportunities and to coordinate across internal business functions.

3002 Strategy costs include the ongoing reporting costs to those entities in scope of internally coordinating, documenting and disclosing climate-related risks and opportunities the company has identified, as well as reporting the impact of these risks on the company's business, strategy, and financial planning. This estimate does not include scenario analysis, which is discussed separately in a later paragraph.

3003 Risk management costs are the ongoing annual costs to those entities in scope to disclose the company's management of climate-related risks, including the coordination across functions internally, identification and assessment of risks and their integration into the company's overarching risk-management strategy. This also includes the time taken to identify and analyze major risk exposures in the context of the company's business strategy.

3004 The FCA presents aggregated costs for governance, strategy, and risk management disclosure instead of individual costs for each of the aforementioned disclosure categories. The cost discrepancy relative to the BEIS rule is primarily driven by significantly different assumptions of internal labor requirements, such as the number of employees, salaries, and required hours.

Another commenter provided cost estimates reported by an anonymous company, referred to as a "Well-known seasoned issuer."[3005] This company, which has made TCFD-aligned disclosures public on its website (including Scope 1, Scope 2, and Scope 3 emissions disclosures) estimated that, combined with the amounts the company currently spends on voluntary climate disclosure, the company would need to spend a total of approximately $35 million over five years to implement climate-related reporting in order to comply with the proposed rules, if adopted as proposed. Within this amount, the company estimates one-time expenses of $19 million and recurring expenses averaging $3.1 million per year. The primary categories of expenses are audit fees, professional services, subscriptions, labor, licenses, and training. The company estimates that compliance with the provisions amending Regulation S-X, as proposed, would have initial costs of $1.5 million to $2.0 million and subsequent ongoing costs of $1.0 million to $2.0 million annually. The company also estimated compliance costs for the proposed Scope 3 emissions disclosure to be $15.6 million over five years, with a significant part of this cost attributable to attestation requirements and with "filing" Scope 3 information.[3006]

Several letters from professional trade or industry organization also provided cost estimates. One commenter stated that the "cost of registrants trying to report in alignment with just certain aspects of TCFD for their first time on a voluntarily basis can be around $500,000…. The actual cost for complete alignment to TCFD could be up to $1,000,000 per registrant over several years," which does not include the annual cost associated with preparing for and conducting attestation.[3007] Another commenter said that based on member feedback, the "true

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[3005] *See* letter from Chamber.

[3006] *See* letter by API. We note that the proposed rules would not have imposed any attestation requirements with respect to Scope 3 emissions disclosure.

[3007] *See* letter from AEPC.

initial set up and ongoing compliance costs for a typical retailer will be more than 35 times the amount that the SEC has estimated....Members estimate that the initial costs of implementing the proposed rules would be somewhere in the $5 million to $15 million range."[3008] Other commenters estimated the initial cost of complying with the proposed amendments to Regulation S-X would exceed $100 million.[3009] One commenter estimated that the combined costs of "complying with the reporting requirements under S-X and S-K would cost companies $3 to $7 million annually."[3010]

One commenter included cost estimates provided by members of its trade association with respect to their on-going efforts, prior to the proposed rules, in measuring GHG emissions.[3011] One member reported that an average automated GHG measuring system would cost $250,000 to purchase and set up, with ongoing annual costs of approximately $100,000. Another member reported that "completing questionnaires and conducting emissions measurements through an automated GHG measuring program with applicable audits costs the company about $15,000 per year to maintain." Another member company's mature Scope 1 and 2 emissions reporting programs resulted in 100 to 200 resource hours per year.

Several individual registrants also provided cost estimates of either their own current climate-reporting practices or expected practices if the proposed rules were adopted as proposed. One multinational registrant that engages in hydrocarbon exploration and production estimated that initial compliance costs with respect to the proposed rules would range from $100 to $500

3008 *See* letter from RILA.

3009 *See* letters from Western Energy Alliance (suggesting initial compliance costs to be "over $100 million for large companies when considering not just the new systems but the staff training required") and API (without specifying whether the $100 million figure reflects implementation costs or ongoing annual costs).

3010 *See* letter from National Retail Federation.

3011 *See* letter from IDFA.

million.[3012] This registrant expressed concerns about the burden of complying with the proposed rules, particularly with the proposed amendments to Regulation S-X. This registrant estimated ongoing costs to be $10 to $25 million annually. One energy company noted that it expected compliance costs to be at least four to five times the estimates provided in the Proposing Release, primarily due to the necessary increases in staff and the added costs in auditing and attestation fees.[3013] Other energy companies estimated that compliance with the proposed amendments to Regulation S-X and reporting Scope 3 emissions would likely exceed $100 million[3014] and $1 million,[3015] respectively.

Another commenter, a multinational energy company, estimated its internal burden hours for Scope 3 emissions reporting to be 650 hours in the first year and 100 hours annually in subsequent years.[3016] A different commenter reported that it allocates one full-time consultant and 20 employees working part time each year from November to March as part of its process to measure Scope 1, Scope 2, and some Scope 3 emissions, collect and validate data, estimate and review emissions, and obtain third-party limited assurance for GHG-related data in its sustainability report.[3017] Another commenter that already tracks some Scope 1 emissions estimated that it may incur an additional cost of $10,556,800 or more to track and report Scope 1 emissions from additional facilities as a result of the proposed rules.[3018] A multinational

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3012 *See* letter from ConocoPhillips.

3013 *See* letter from PPL Corporation.

3014 *See* letter from Western Energy Alliance.

3015 *See* letter from Williams, Inc.

3016 *See* letter from Ørsted.

3017 *See* letter from Air Products and Chemicals, Inc.

3018 *See* letter from Energy Transfer LP. This commenter derived this cost based on estimates from the EPA's mandatory GHG reporting rule. *See Mandatory Reporting of Greenhouse Gases*, 74 Fed. Reg. 56, 260, 363 tbl. VII-2 (Oct. 30, 2009). This commenter estimated this cost to be $7,000,000 in 2006 dollars, which was adjusted for inflation to obtain $10,556,800 in 2023 dollars.

fertilizer company estimated that the direct and indirect costs of compliance with the proposed rules would be between $35 million and $55 million, with assurance costs related to financial statement metrics estimated to be $70,000 to $225,000 annually.[3019]

We also reviewed memoranda of staff meetings with external parties that further inform our assessment of the final rules' compliance costs.[3020] One organization presented pricing information for the following relevant services provided: TCFD reporting, excluding measuring emissions and establishing targets ($100,000 average); assessing Scopes 1, 2, and 3 emissions ($75,000 to $125,000); and target setting ($20,000 to $30,000).[3021] A different organization indicated fees would range from $11,000 to $105,000 for services related to GHG accounting (Scopes 1, 2, and 3 emissions).[3022] Another organization estimated that costs for assessing Scopes 1 and 2 emissions would range between $25,000 and $45,000 and assessing Scopes 1, 2, and 3 emissions would cost between $50,000 and $125,000, depending on whether a given company already has emissions-measuring systems and processes in place.[3023]

3019 *See* letter from Nutrien. This estimate includes costs associated with conducting scenario analysis and including the related information in public disclosures; measuring and reporting Scope 1 and 2 emissions by each GHG, obtaining reasonable assurance on Scope 1 and 2 emissions by each GHG; measuring and reporting Scope 3 emissions by each GHG for public disclosure subject to DCP; and disclosure of the proposed Financial Impact Metrics within the audited financial statements, among other proposed disclosures. These costs include internal costs, external professional service fees, and additional systems and internal control processes that the commenter indicated would need to be designed and operating effectively for public disclosure of high-quality information.

3020 The meeting memoranda are available at the same location as the comment letters in response to the Proposing Release. *See supra* note 19. Some of these meetings occurred prior to the Proposing Release and thus any data included in the memoranda do not reflect specific details of the proposed rules; however, we have considered these memoranda as part of this assessment as they contain relevant cost information.

3021 *See* Memorandum Concerning Staff Meeting With Representatives of S&P Global (Feb. 4, 2022); *see also* Proposing Release, *supra* note 1027.

3022 *See* South Pole Memo; *see also* Proposing Release, note 1037. These numbers have been converted from EUR based on the 2022 average exchange rate of $1.0538 USD/EUR, rounded to the nearest $100.

3023 *See* Memorandum Concerning Staff Meeting with Representatives of Persefoni (Nov. 30, 2021); *see also* Proposing Release, at n. 1036.

The cost information in the above sources indicates the variance and the scale of compliance costs the proposed rules would have imposed on registrants. We note, however, that many of the estimates combine the costs of multiple components without providing a breakdown of component costs, which makes it difficult to isolate only the components that are applicable to the proposed or final rules. Furthermore, these voluntary cost estimates may reflect some selection bias such that they may be skewed toward a certain demographic (e.g., large-cap companies) and thus may not be representative of the broad sample of affected registrants. Finally, to the extent that the cost estimates are specific to the proposed rules, they do not account for the changes made to the final rules. For example, the final rules' requirements with respect to financial statements have been narrowed relative to the proposed rules.[3024] In addition, the final rules do not require the disclosure of Scope 3 emissions. Nevertheless, we use this cost information to the extent possible to inform our assessment of the expected compliance costs of the final rules, as outlined in the following subsection.

b. Direct cost estimates for the final rules

The final rules will impose a number of new disclosure requirements on registrants. These requirements will result in additional compliance costs for registrants, and, depending on the nature of the registrant's operations and its existing disclosure practices, these additional compliance costs could be significant. Using comment letters and other sources, we take a conservative approach (i.e., erring on the side of overstating costs rather than understating them) to estimate approximate compliance costs for the final rules, which are discussed in subsequent sections and summarized immediately below.

[3024] *See*, *e.g.*, *supra* sections II.K.2.c and II.K.3.c.

With respect to the Regulation S-K amendments pertaining to governance disclosure (Item 1501); disclosure regarding the impacts of climate-related risks on strategy, business model, and outlook (Items 1502(a) through (e) and (g)); and risk management disclosure (Item 1503), we estimate that compliance costs will be $327,000 in the first year of compliance and $183,000 annually in subsequent years.[3025] For those registrants that conduct scenario analysis and are required to provide attendant disclosures (Item 1502(f)), we estimate the reporting costs will be $12,000 in the first year and $6,000 in subsequent years.[3026] Some registrants will be required to disclose Scope 1 and 2 GHG emissions (Item 1505) after a specified phase in period. We estimate that the compliance costs for these disclosures will be $151,000 in the first year of compliance and $67,000 annually in subsequent years.[3027] After an additional phase in period, applicable registrants will be required to obtain assurance for their emissions disclosures (Item 1506). Limited assurance for emissions disclosures is estimated to cost $50,000 while reasonable assurance is estimated to cost $150,000.[3028] For registrants that voluntarily establish targets or goals and are required to provide attendant disclosures (Item 1504), we estimate the reporting costs will be $10,000 in the first year of establishing the target and $5,000 in subsequent years.[3029] With respect to amendments to Regulation S-X, we estimate an upper bound of $500,000 in the first year of compliance, while the annual cost in subsequent years is

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[3025] *See* section IV.C.3.b.i.

[3026] *See* section IV.C.3.b.iv.

[3027] *See* section IV.C.3.b.ii.

[3028] *See* section IV.C.3.b.iii.

[3029] *See* section IV.C.3.b.iv.